

Original article

Cardioprotection of Sheng Mai Yin a classic formula on adriamycin induced myocardial injury in Wistar rats

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ABSTRACT

Background: Sheng Mai Yin (SMY), a well-known Chinese herbal medicine, is widely used to treat cardiac diseases characterized by the deficiency of Qi and Yin syndrome in China. SMY-based treatment has been derived from Traditional Chinese Medicine (TCM), officially recorded in the Chinese Pharmacopoeia.

Purpose: We aimed to clarify whether SMY attenuates myocardial injury induced by adriamycin in Wistar rats with chronic heart failure (CHF).

Methods: To quantify ginsenoside Rg1, ophiopogonin D, ophiopogonin D', schisandrin by HPLC. To establish CHF animal model, adriamycin was intraperitoneally injected in Wistar rats for 7 weeks at a dose of 2 mg/kg body weight. Overall, 180 rats were randomly assigned to six groups: control, CHF model, captopril (positive control), high dose (HSMY), medium dose (MSMY), and low dose (LSMY). Experimental rats were fed 0.625 mg/kg captopril and 90 mg/kg, 45 mg/kg, and 22.5 mg/kg SMY, respectively, over 7 weeks. The inflammatory cytokines TNF- α and IL-6 were measured using ELISA. Matrix metalloproteinases (MMPs) were identified using immunohistochemistry (IHC). Both IHC and RT-PCR were used for quantification of COL-IV expression levels in the heart tissues. Scanning electron microscopy (SEM) was used for the visualization of myocardium morphology.

Results: The concentration of ginsenoside Rg1, ophiopogonin D, ophiopogonin D' and schisandrin in SMY was found to be 25.63 ± 3.42 mg, 11.00 ± 1.17 mg, 7.02 ± 0.51 mg, and 25.31 ± 4.28 mg per gram of SMY, respectively. Compared with CHF model group, TNF- α levels were significantly lower ($p < .01$) in the four drug-administered groups. Moreover, except in the SYM low dose group, IL-6 levels in the other 3 drug-administered groups were also significantly reduced ($p < .01$). COL-IV expression was also significantly reduced on treatment with high SYM dose ($p < .05$). IHC results confirmed that SMY and captopril significantly reduced MMPs expression in the heart.

Conclusion: SMY could control or slow CHF progression by suppressing pathological changes in the myocardium in CHF models. This could be attributed at least partly to the downregulation of IL-6 and TNF- α and inhibition of overexpression of MMPs and COL-IV, which significantly relieved the cardiac-linked pathologies, decreased the risk of myocardial fibrosis, and inhibited cardiac remodeling. These findings suggested that SMY and captopril have similar efficacy for the treatment of adriamycin-induced myocardial injury. In addition, Chinese herbal preparation SMY may play a role in the treatment of cardiac diseases.

Introduction

Natural products possess an enormous structural and chemical diversity, which cannot be matched by synthetic libraries of small

molecules and thus continue to inspire novel discoveries in chemistry, biology, and medicine. These natural chemicals have been evolutionarily optimized as drug-like molecules and remain the best known sources of drugs and drug-leads (Newman and Cragg, 2012). In the last

Abbreviations: CA, Captopril administration group; CHF, Chronic heart failure; CHFM, CHF model group; COL-IV, Collagen-IV; CON, Control group; ECM, Extracellular matrix; ELISA, Enzyme-linked immunosorbent assay; HF, Heart failure; HSMY, High dose Sheng Mai Yin administration group; IHC, Immunohistochemistry; IL-6, Interleukin-6; LSMY, Low dose Sheng Mai Yin administration group; MI, Myocardial infarction; MMP-2, Matrixmetalloproteinases-2; MMP-9, Matrixmetalloproteinases-9; MMPs, Matrix metalloproteinases; MSMY, Middle dose Sheng Mai Yin administration group; SEM, Scanning electron microscopy; SMY, Sheng Mai Yin; TCM, Traditional Chinese Medicine; TNF- α , Tumor necrosis factor- α

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decade, de-replication has emerged as a hot research topic, leading to a huge publication boom since 2012. This blending of multiple disciplines in ways that provide important and novel conceptual and/or methodological advances has opened up vast research prospects (Gaudêncio and Pereira, 2015; Leung et al., 2011). This interdisciplinary research has also led to the identification of natural products like 6,6"-biapiogenin, which is only the second inhibitor discovered so far for NEDD8-activating enzyme. Importantly, 6,6"-biapiogenin was found to be enzymatically active in kinetics- and cell-based assays, with a potency in the micromolar range. Fong et al. (2007) showed that the extract of the rhizomes of *Alisma orientalis* (Sam) Juzep. has synergistic growth inhibitory effect with cancer drugs that are P-glycoprotein substrates including actinomycin D, puromycin, paclitaxel, vinblastine, and doxorubicin. Zhong et al. (2015) firstly reported that natural product-like compound 1 was the first natural product-like inhibitor and only the second inhibitor overall of TLR1-TLR2 heterodimerization, as potential agents for the treatment of inflammatory and autoimmune diseases. Amentoflavone was found to be JAK2 inhibitor by structure-based virtual screening of a natural product library, and its analogues might function as Type II inhibitor of JAK2 (Ma et al., 2014). Liu et al. (2014) indicated that natural product-like compound 1 inhibited STAT3 DNA-binding activity *in vitro* and attenuated STATA3-directed transcription in *cellulo* with selectivity over STAT1 and with comparable potency to the well-known STAT3 inhibitor S31-201, and also exhibited selective anti-proliferative activity against cancer cells over normal cells. Peptidyl-proline isomerases (PPIases) played a key role in cancer, neurodegeneration, and psychiatric disorders, however, macrocyclic natural products might create potent and selective inhibitors, such as FK506, rapamycin, and cyclosporin. Manivannan et al. (2017) illustrated that twelve novel silybin analogues had significantly greater efficacy than silybin, and derivative 15k as a novel tubulin inhibitor with significant activity against ovarian cancer cells. Nature has been extremely generous to the mankind in offering life-saving therapies, and the next great drug may be just around the corner: are we ready to seize the opportunity? (Shen, 2015).

Sheng Mai Yin (SMY) is a classical natural and effective formula, which is routinely used in China, and contains Radix Ginseng (*Panax ginseng* C.A. Mey., Araliaceae), Radix Ophiopogon (*Ophiopogon japonicus* (Thunb.) Ker-Gawl., Liliaceae) and Fructus Schisandrae (*Schisandrae chinensis* (Turcz.) Bail., Magnoliaceae) (Chen et al., 2007; Wang et al., 2005; Committee of Pharmacopoeia of PR China, 2005). "Sheng" and "Mai" are the Chinese abbreviations for *Panax ginseng* and *Ophiopogon japonicus*, which have both been extensively used for centuries in China as effective drugs (Chen et al., 2007; Gillis, 1997). Medicinal plants have been used in patients with congestive heart failure as well as systolic hypertension (Rastogi et al., 2016). Specifically, *Panax ginseng*, *Fructus Schisandrae*, and *Ophiopogon japonicus*, which contain multiple bioactive components, have been shown to be effective against many diseases (Wang et al., 2005; Zhang et al., 2010). Given the excellent activity and safety of its components, SMY is widely used for the treatment of cardiac diseases, which are characterized by deficiency of Qi and Yin syndrome (Mo et al., 2015). Clinically, SMY has been shown to treat shock, coronary heart disease, angina, myocardial infarction (MI), viral myocarditis, pulmonary heart disease, heart failure etc. In addition, SMY can treat myocardial diseases, rheumatoid diseases, systemic lupus erythematosus, as well as epidemic hemorrhagic fever. These diverse curative effects of SMY explain why SMY in combination with other herbs has been used for the treatment of diabetes, nodular lupus erythematosus, mild brain dysfunction syndrome, optic atrophy, recurrent pneumothorax, iron deficiency anemia, severe infectious mononucleosis, and malignant tumor (with a maximum clinical dosage of 0.3 g/kg) (Zhang et al., 2010). Interestingly, SMY is also especially prescribed for coronary artery disease (Wang et al., 2002).

On the cellular level, SMY suppresses mitochondrial apoptosis as indicated by reduction in several pro-apoptotic factors (Bax,

Table 1

Primers (F, forward; R, reverse) used for relative mRNA quantifications by real-time PCR.

Target	Primer sequence(5'–3')	Length (pb)
COL-IV	F: GAGGGTGTGGACAAGC	5300.50
	R: TAAATGGAGCTGGCTCGGAATTC	6774.48
GAPDH	F: TGTGAGGAGATGCTCAGTG	6253.13
	R: GGCATTGCTCTCAATGACAA	6101.04

cytochrome c, and cleaved caspase-3) and up-regulation of the anti-apoptosis factor Bcl-2 (Mo et al., 2015). The improvement in cardiac contractile function afforded by SMY treatment is likely mediated by an increase in Ca^{2+} release from SR through L-type Ca^{2+} current-activated RyRs (Zhang et al., 2008). On the other hand, SMY improves the post-ischemic myocardial dysfunction by opening the mitochondrial K_{ATP} channels (Wang et al., 2002), improving the heart structure and reducing Cx43 expression after MI. SMY also inhibits myocardial fibrosis in rats with diabetic cardiomyopathy, and significantly delays the formation of diabetic cardiomyopathy through multiple signaling pathways (Ni et al., 2011). In view of these considerations, the aim of our study was to investigate the basis of the protective function of SMY on myocardial injury and on the regulation of IL-6 and TNF- α levels. Our study also explores the regulatory role of SMY on MMPs and COL-IV to achieve myocardial remodeling as well as the protective function of SMY against the pathological changes of the myocardium.

Materials and methods

Preparation of SMY

SMY was purchased from Chiatai Qingchunbao Pharmaceutical Co. (Zhejiang, China) (Batch no.- 1410014). This SMY preparation was a 1320 g mixture of three common Chinese herbal medicines: *Ginseng radix*, *Ophiopogonis Radix* and *Schisandrae Chinensis Fructus* mixed in a ratio of 1:2:1.

Reagents and drugs

Adriamycin was purchased from Shenzhen Main Luck Pharmaceuticals Inc. (Shenzhen, China). Captopril was purchased from North China Phar. Co. (Hebei, China). Rat IL-6, TNF- α enzyme-linked immunosorbent assay (ELISA) assay kits were obtained from RayBiotech. Inc. (GA, USA). Histostain-Plus kits was purchased from ZSBO. Inc. (Beijing, China). Antibodies for type IV collagen, matrixmetalloproteinases-2 and matrixmetalloproteinases-9 were purchased from Boster Co. (Beijing, China). All other agents used in this study were of commercially available grade and purity.

Animals and CHF model

Adult Wistar rats weighing 160–200 g, with an equal proportion of males and females, were provided by the Animal Breeding Center of Lanzhou Military Region General Hospital. These animals were housed under controlled conditions at a temperature of $25 \pm 2^\circ\text{C}$, humidity of $40 \pm 5\%$ and on a 12 h light-dark cycle. The rats had free access to solid rodent chow and tap water. Animals were allowed a 1 week acclimatization period prior to entry into any experimental protocol. The entire laboratory procedure was carried out under the permission and surveillance of local ethics committee. The experimental procedures were approved by Lanzhou Institute of Husbandry and Pharmaceutical Sciences, CAAS.

Adriamycin was used to establish the animal model of CHF. It was injected intraperitoneally for 7 weeks at a dose of 2 mg/kg body weight (Li et al., 2006; Cheng, 2011). The rats were randomly assigned into six groups. Group 1 (CON) was normal controls comprising of 30 healthy

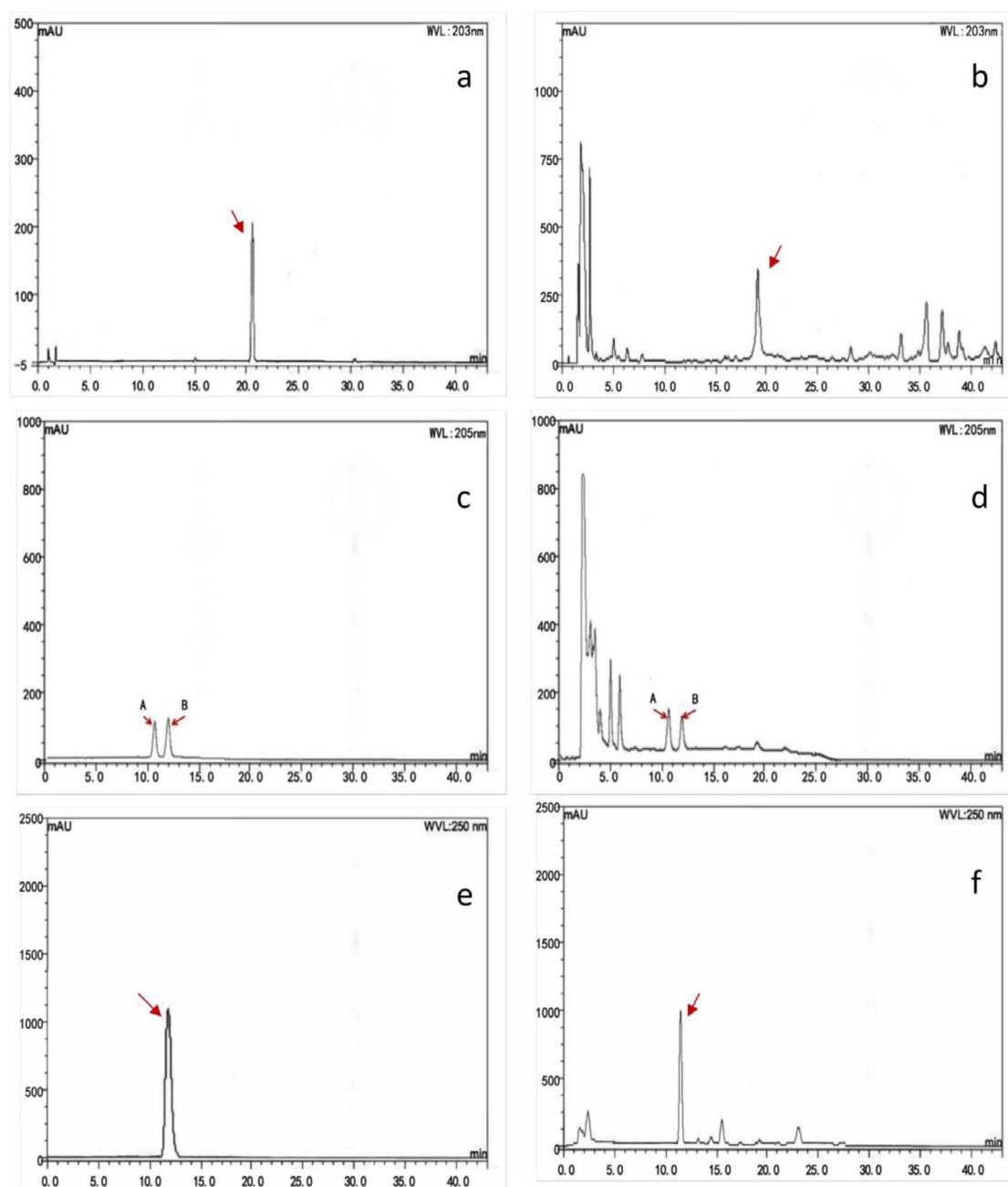


Fig. 1. HPLC chromatogram map. (a) reference substance (ginsenoside Rg1, 203 nm), (b) SMY test sample (203 nm), (c) reference substance (A- ophiopogonin D, B- ophiopogonin D', 205 nm), (d) SMY test sample (205 nm), (e) reference substance (schisandrin, 250 nm), (f) SMY test sample (250 nm).

rats fed under the same conditions as the experimental group but lacking any treatment. Group 2 (CHF model control, CHFM) were rats, which were not subjected to any treatment after the establishment of CHF in them. Group 3 (CA, positive control) were animals treated with captopril at a dosage of 0.625 mg/kg body weight. Group 4 (HSMY) were treated with high dosage of SMY (90 mg/kg body weight per day) for 7 weeks after the establishment of CHF model. Group 5 (MSMY) were treated with medium dosage of SMY (45 mg/kg body weight per day) for 7 weeks after the establishment of CHF model. Group 6 (LSMY) consisted of animals treated with low dosage of SMY (22.5 mg/kg body weight per day) for 7 weeks after the establishment of CHF model. At the end of the protocol, cardiac function was examined physiologically and heart tissues were harvested for performing IHC and biochemical analyses.

Quantification of ginsenoside Rg1, ophiopogonin D, ophiopogonin D', schisandrin by HPLC

Precisely weighed samples (ginsenoside Rg1, ophiopogonin D, ophiopogonin D', schisandrin and SMY test sample) were extracted with methanol in an ultrasonic bath and filtered in a 0.45 μ m filter. An aliquot of 20 μ l of each sample was injected onto the HPLC column (kromasil100-5C₁₈ 250 $\times$ 4.6 mm, 5 μ m particle size) and elution was carried out with acetonitrile: water (ginsenoside Rg1 6:4, ophiopogonin D 5.5:4.5, ophiopogonin D' 5.5:4.5, schisandrin 1:1) at a flow-rate of 1 ml/min and eluate was monitored at 203 nm (ginsenoside Rg1), 205 nm (ophiopogonin D, ophiopogonin D') and 250 nm (schisandrin) (Committee of Pharmacopoeia of PR China, 2005). The procedure was repeated three times for each sample. Each solution was prepared and injected three times and the curve was plotted using an average of the area. The calculated concentrations of ginsenoside Rg1, ophiopogonin

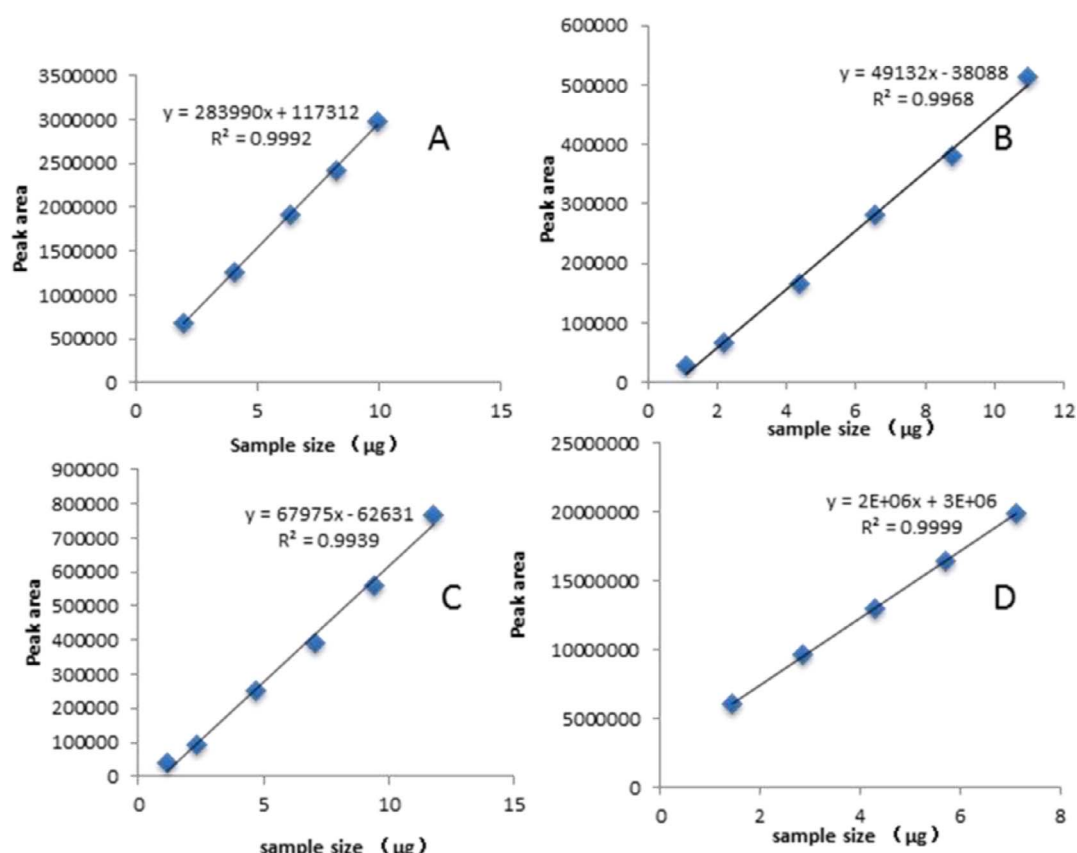


Fig. 2. Calibration curve of standard substances obtained using HPLC. (A) ginsenoside Rg1, (B) ophiopogonin D, (C) ophiopogonin D', (D) schisandrin.

D, and ophiopogonin D' were expressed in terms of mean $\pm$ standard deviation (SD) (mg/g).

Determination of TNF- α and IL-6 levels in heart tissues

The inflammatory cytokines TNF- α and IL-6 were quantified using ELISA kits according to the manufacturer's instructions. Myocardial tissues were homogenized in RIPA lysis buffer, and centrifuged at 6000 g for 15 min at 4 °C. 0.1 ml supernatant was removed and mixed with 1 ml normal saline. The supernatants were collected for the analyses, and data were expressed as picograms per milligram of protein.

Real-time PCR (q-PCR)

We quantified the mRNA expression levels of COL-IV and GAPDH (reference) by One Step STBR® PrimerScript® RT-PCR Kit (TaKaRa, Japan) using the standard curve method (Terova et al., 2011). Primer 5.0 software was used to design the primers for COL-IV and GAPDH (Zhao et al., 2014). The nucleotide sequences of all primers used in this study are reported in Table 1. Real-time analysis was performed in duplicate for each sample using SYBR® Premix Ex Taq™ II (Tli RNaseH Plus, TaKaRa, Japan). The following real-time run conditions were used: 2 min at 50 °C, 30 min at 60 °C and 5 min at 95 °C, followed by 35 cycles consisting of 20 s at 94 °C, 30 s at 63 °C, and 30 s at 72 °C. We used iQ5 Real-Time PCR system (Bio-Rad, USA) to perform SYBR® Premix Ex Taq™ II reactions and collected runs data using the iQ5™ software (Bio-Rad, USA). Cycle threshold (Ct) value obtained by each standard mRNA amplification was used to create a standard curve for each target gene. This curve served as a basis for calculating the unknown mRNA levels of each gene present in the total RNA extracted from each sample.

Myocardial sample preparation for scanning electron microscopy (SEM)

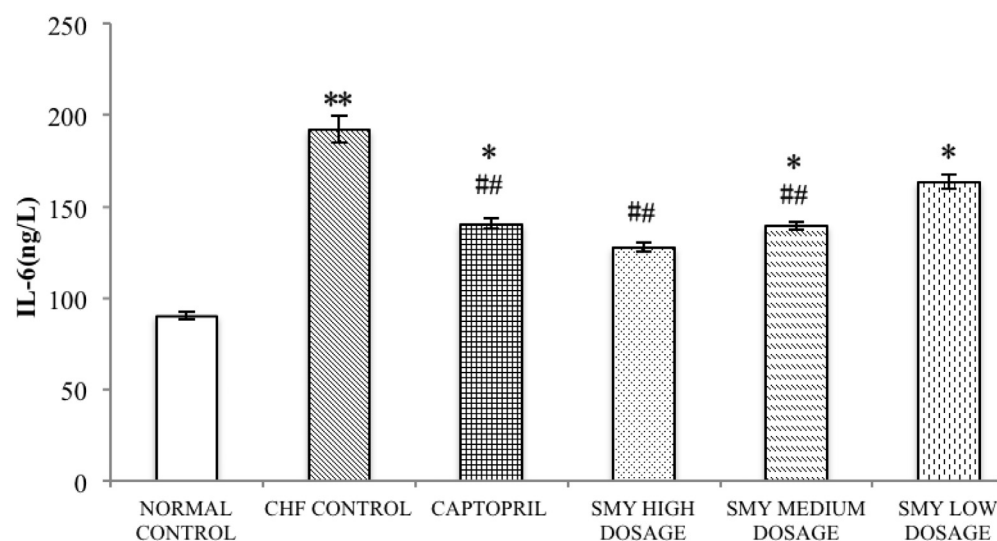
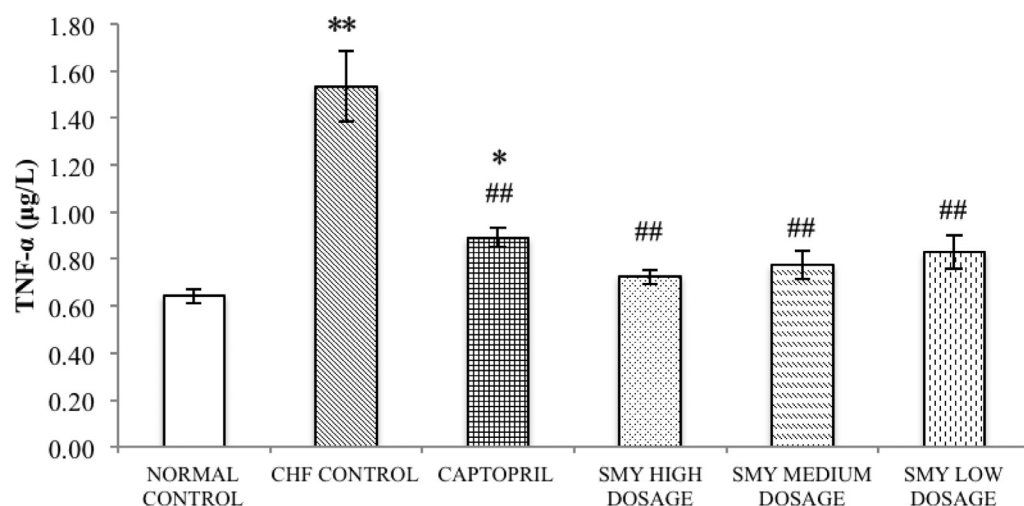
Using a scanning electron microscope JSM-6510A SEM (JEOL, Japan), the diameter of myocardial fibers was measured using the Smile Shot™ system software. For sample preparation, the following procedure has been proposed (Li, 2008). Cardiac tissue was first washed using Phosphate Buffer Solution (PBS) buffer before being refrigerated in 2–4% glutaraldehyde at 2–4 °C. The treated cardiac tissue was then rinsed with double-distilled water, and washed again with PBS. Ethyl alcohol substitution and dehydration was performed in a series from 40% $\rightarrow$ 50% $\rightarrow$ 60% $\rightarrow$ 70% $\rightarrow$ 80% $\rightarrow$ 90% $\rightarrow$ 95% $\rightarrow$ 100% with each dehydration step being about 15 min long. 100% ethyl alcohol substitution was done 3 times before the treated tissue was immersed in tert-butyl alcohol 2 h to substitute ethyl alcohol. Finally, vacuum freezing drying and gold coating were performed on the dehydrated and tert-butyl alcohol-substituted samples.

Immunohistochemistry

IHC staining for type IV collagen (COL-IV 1:1000, BOSTER, China), matrixmetalloproteinases-2 (MMP-2 1:500, BOSTER, China) and matrixmetalloproteinases-9 (MMP-9 1:250, BOSTER, China) were performed using the standard streptavidin-biotin-peroxidase on 5 μ m thick sections of formalin-fixed, paraffin-embedded tissue. The sections were de-paraffinized in xylene, and rehydrated through graded alcohols to distilled water before antigen retrieval by heat method in citrate solution (pH 6.0). An automated detection using Leica ST5010 autostainer XL (Lanzhou, China) was utilized for analysis.

Statistical analysis

Data were expressed as mean $\pm$ SD. The effect of treatments on the expression of biotransformation genes were made by one-way variance



COL-IV expression in heart

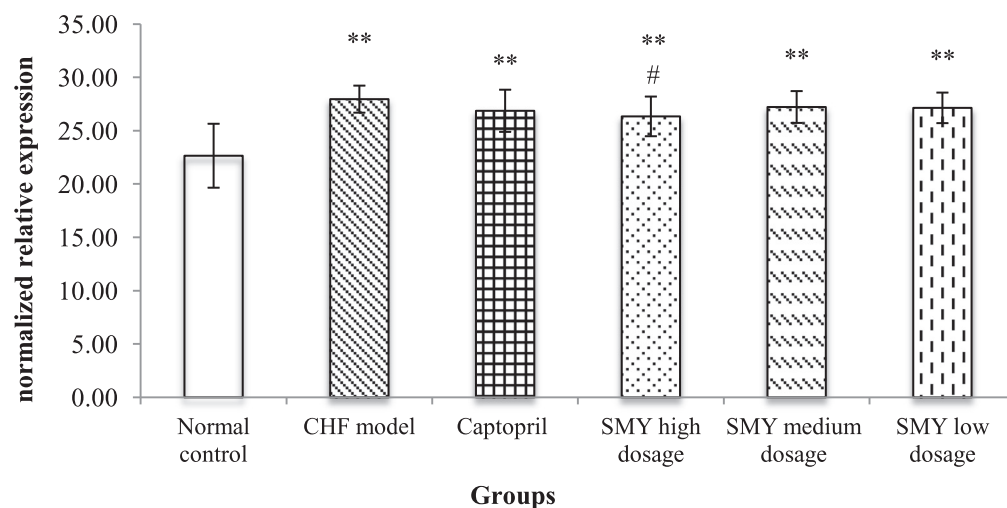


Fig. 3. Effect of SMY on TNF- α and IL-6 among six groups (Mean $\pm$ SD). TNF- α and IL-6 of CHFM ($p < .01$) and CA ($p < .05$) were significantly increased, IL-6 of MSMY and LSMY was significantly increased ($p < .05$), respected to CON, * indicates significant differences ($p < .05$), ** indicates extremely significant differences ($p < .01$). TNF- α of 4 drug administration groups were extremely significant decreasing ($p < .01$). Beside of LSMY, IL-6 of the other three drug administration groups were extremely significant decreasing ($p < .01$), respected to CHFM, # indicates significant differences ($p < .05$), ## indicates extremely significant differences ($p < .01$).

Fig. 4. Expressions of COL-IV in heart tissues (Mean $\pm$ SD). Data were analyzed by one-way ANOVA, with $p < .05$ indicating statistical significance. All 5 groups significantly increased COL-IV in heart tissues. CON was the reference group in pairwise comparison. * indicated $p < .05$, ** indicated $p < .01$. HSMY significantly decreased COL-IV in heart tissues, CHF was the reference group in pairwise comparison. # indicated $p < .05$, ## indicated $p < .01$.

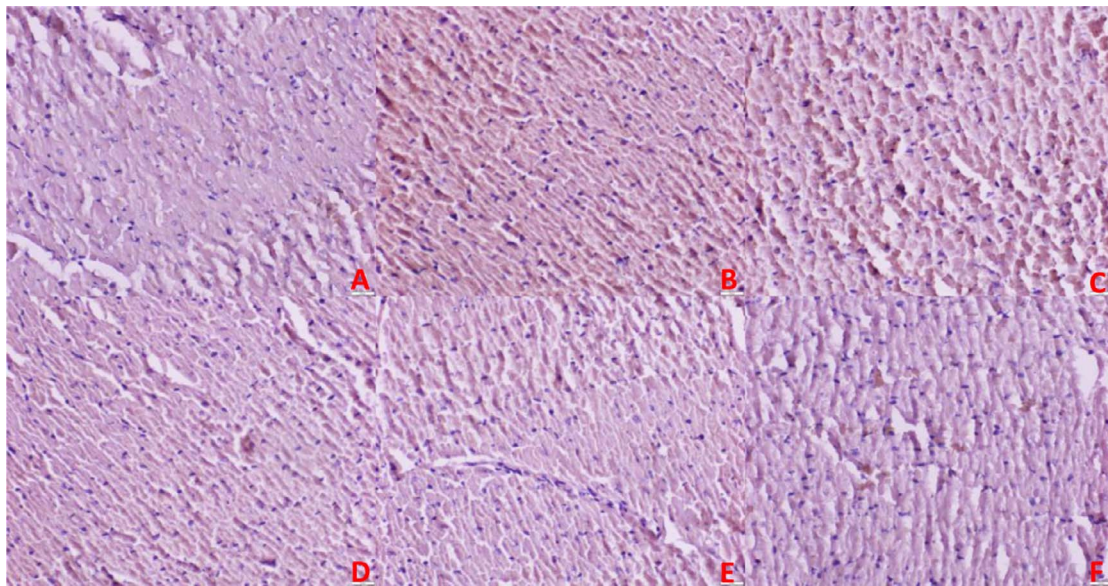


Fig. 5. Results of COL-IV expression by IHC in heart tissues. (A) The expression quantity of COL-IV was below 20%, it could be considered the negative sample (Negative, CON, $\times 400$). (B) CHFM ($\times 400$). (C) CA ($\times 400$). (D) HSMY ($\times 400$). (E) MSMY ($\times 400$). (F) LSMY ($\times 400$). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

analysis (ANOVA) test using the Bonferroni post-hoc, followed by Student–Newman–Keuls pair-wise test, taking $p = .05$ as a significant cut-off. All calculations were performed using GraphPad Prism® 5 software.

Results

Quantitative analysis of ginsenoside Rg1, ophiopogonin D, ophiopogonin D', schisandrin

The chromatogram obtained for ginsenoside Rg1, ophiopogonin D, ophiopogonin D', schisandrin (Fig. 1) showed distinct peaks for each of them. The calibration curves (Fig. 2) were prepared using standard substances and were found to be linear in the concentration range used ($r^2 = 0.99$). The concentration of ginsenoside Rg1, ophiopogonin D, ophiopogonin D' and schisandrin in SMY was found to be 25.63 ± 3.42 mg, 11.00 ± 1.17 mg, 7.02 ± 0.51 mg, and 25.31 ± 4.28 mg per gram of SMY, respectively.

Effects of SMY on inflammatory cytokine levels in rats with CHF

In comparison to CON (Fig. 3), the levels of both TNF- α and IL-6 in CHF rats were significantly higher ($p < .01$). There was a significant difference in the levels of TNF- α and IL-6 ($p < .05$) in CA treated animals compared to controls. IL-6 levels in MSMY and LSMY were also significantly higher ($p < .05$). Overall, in comparison with CHFM, TNF- α levels in all of the four drug-administered groups were significantly lower ($p < .01$). Except for LSMY, IL-6 levels for the three other drug-administered groups were significantly lower ($p < .01$). Among the four drug administration conditions, no significant difference was observed ($p > .05$), however as the dose of SYM decreased, the levels of TNF- α and IL-6 showed an increasing trend ($p > .05$).

Effects of SMY on COL-IV

In comparison to CON (Fig. 4), all of the other 5 groups significantly increased COL-IV expression levels in heart tissues ($p < .01$). In comparison with CHFM, the COL-IV expression of HSMY was significantly reduced ($p < .05$) in heart tissues (Fig. 5). Among CA and the 3 SMY-treated groups, there was no significant difference in COL-IV

levels ($p > .05$) (Fig. 5). The levels of COL-IV in CHFM were considerably higher than those in the control group, indicating an over-production of extracellular matrix in model rats (Fig. 5).

Effects of SMY on MMPs

MMP-2 and MMP-9, two major indicators of ventricular remodeling were also detected by IHC to further evaluate myocardial fibrosis (Figs. 6 and 7). The expression level of MMP-2 and MMP-9 was significantly increased in the model CHFM compared to the level in control group. CA and SMY treatment groups reduced their expressions back toward normal level.

Scanning electron microscopy

In the CON, the myocardium fiber was integrated without any degeneration, necrosis and fibrosis, with a diameter of $8.73\text{--}10.15\text{ }\mu\text{m}$ (Figs. 8 and 9). Drug-treatment in the five other groups presented different degree of degeneration, fibrosis or both. In CHFM, the myocardium fiber was not integrated, and there was significant degeneration, the myocardial diameter was only $3.52 \pm 0.51\text{ }\mu\text{m}$. In CA, fibrosis was not presented, but the myocardium fiber was incomplete with ruptured fragments, and the myocardial diameter was $3.99 \pm 0.38\text{ }\mu\text{m}$. In the three SMY-treated groups, HSMY showed an integrated myocardium fiber without any significant degeneration, however slight fibrosis was still observed. Both MSMY and LSMY presented light fibrosis but the completeness of myocardial fiber was higher than in the CHFM without any significantly ruptured fragments. Overall, with a decreasing dosage of SMY, the myocardial diameters showed a decreasing trend.

Discussion

Several active components such as schisandrol A, which is one of the key active components of SMY, have been shown to be bioactive in vivo. It has been shown that these natural compounds activate eNOS and expression of Bcl-2, and reduce collagen deposition and expression of Bax and ASK1 in myocardial tissues. They also inhibit caspase-3 activity and mitochondrial permeability in H9c2 cells (Wang et al., 2010; Park et al., 2012; Chen et al., 2013; Chun et al., 2014; Chiu et al.,

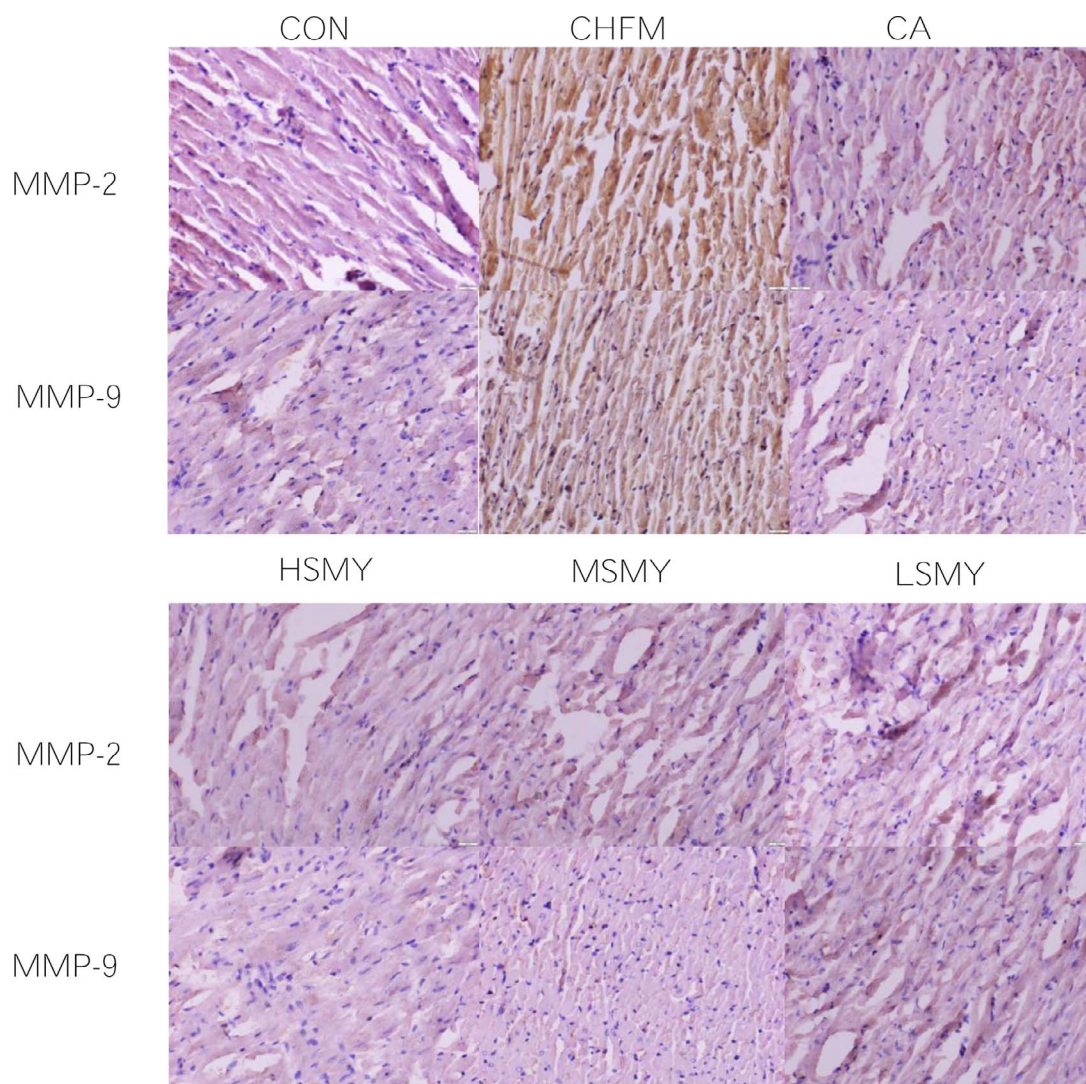


Fig. 6. Results of MMPs expression in heart tissues. (A) The expression quantity of MMP-2 was below 20%, it could be considered the negative sample (Negative, CON, $\times 400$). (B) CHFM ($\times 400$). (C) CA ($\times 400$). (D) HSMY ($\times 400$). (E) MSMY ($\times 400$). (F) LSMY ($\times 400$). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

2008). Schisandrin B has also been shown to prevent doxorubicin-induced cardiac dysfunction by modulation of DNA damage, oxidative stress and inflammation through inhibition of MAPK/p53 signaling (Thandavarayan et al., 2015). Eight ginsenoside monomers Rb1, Rg1, Rf, Rh1, Rc, Rb2, Ro, and Rg3 have been reported to act as NF- κ B inhibitors, thereby they can downregulate TNF- α , IL-6, IL-8 levels through the inhibition of NF- κ B (Xing et al., 2013). The cellular response to IL-6 in the heart has also been well characterized. Cardiac tissues provide revealing examples of how the duration of IL-6 signaling relates to the chronicity of the disease and demonstrates the transition from protective to pathogenic. Cardiac myocytes themselves make IL-6 in response to injury and in addition to an increase in IL-6 signaling, increased IL-6 production is also associated with depressed cardiac function (Yang et al., 2004). Increased IL-6 plays a role in late phase preconditioning that confers cardio protection (Dawn et al., 2004; Smart et al., 2006). However, chronic elevated myocardial production of IL-6 family cytokine, which occurs post-MI and in HF, has been associated with worsening of heart outcomes (Terrell et al., 2006; Wollert and Drexler, 2001). SHR-developed cardiac hypertrophy complicated with diastolic heart dysfunction, increased expression of brain natriuretic peptide, downregulation of beta adrenergic receptors and simultaneous up-regulation of IL-6, which indicates active proinflammatory process, at least partly, explain the pathologies during early stage when cardiac

hypertrophy associated with diastolic dysfunction occurs (Haugen et al., 2007). According to Chinese Pharmacopoeia, ginsenoside Rg1 was used to control the quantification standards of SMY (Committee of Pharmacopoeia of PR China, 2005). However, SMY contained Ginseng, Ophiopogon, Schisandrae, based on the related literature, ginsenoside Rg1, ophiopogonin D, ophiopogonin D' and schisandrin have various bioactivities on cardiac diseases. For instance, ginsenoside Rg1 showed that protect cardiomyocytes under hypoxic conditions by reducing intracellular Ca^{2+} overload (He et al., 2014). Ophiopogonin D and ophiopogonin D' showed those attenuate doxorubicin induced autophagic cell death by relieving mitochondrial damage in vivo and in vitro (Zhang et al., 2015). Schisandrin showed that enhance glutathione antioxidant response in H9c2 cells (Ko and Chiu, 2005), prevent doxorubicin induced cardiac dysfunction by modulation of DNA damage, oxidative stress and inflammation through inhibition of MAPK/p53 signaling (Thandavarayan et al., 2015), and inhibit NADPH-dependent and CYP450-catalyzed reaction in the myocardial tissues, and enhance glutathione antioxidant response in H9c2 cells (Chen and Ko, 2010). In present study, ginsenoside Rg1, ophiopogonin D, ophiopogonin D', schisandrin in SMY were used to be the quantification standards of SMY, that was not merely quantification standards of SMY, but also a better control of CHF progression. Quantification standards of formula would provide the scientific basis for the further

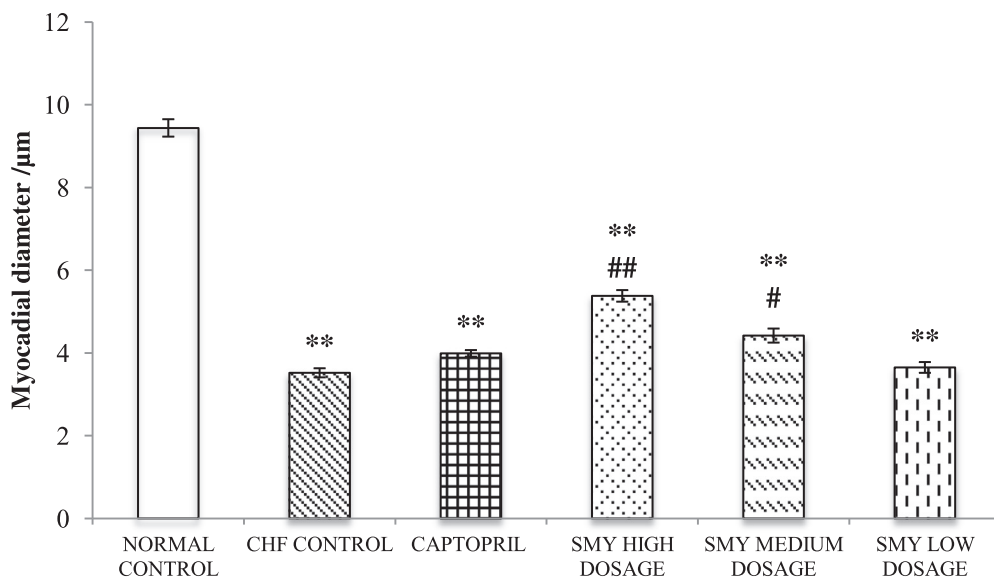
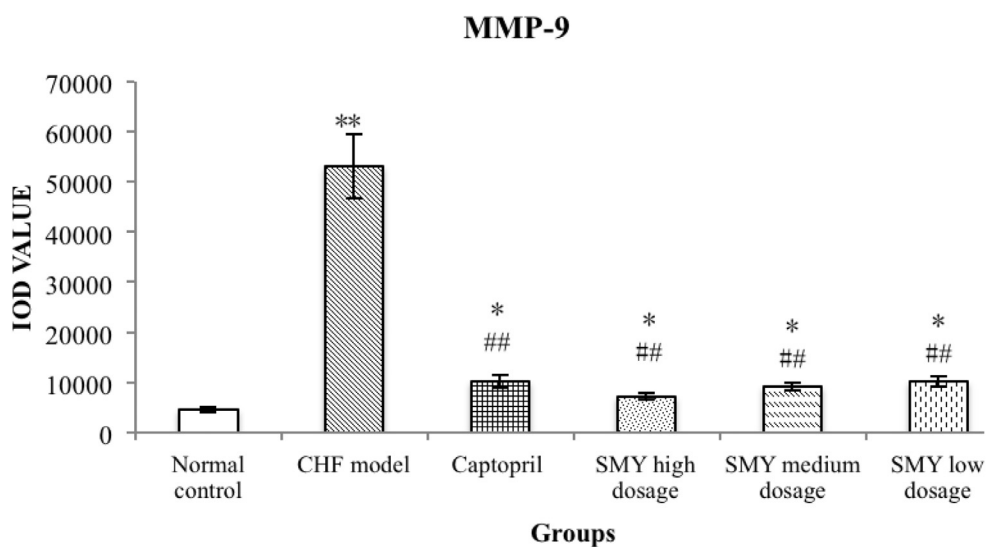
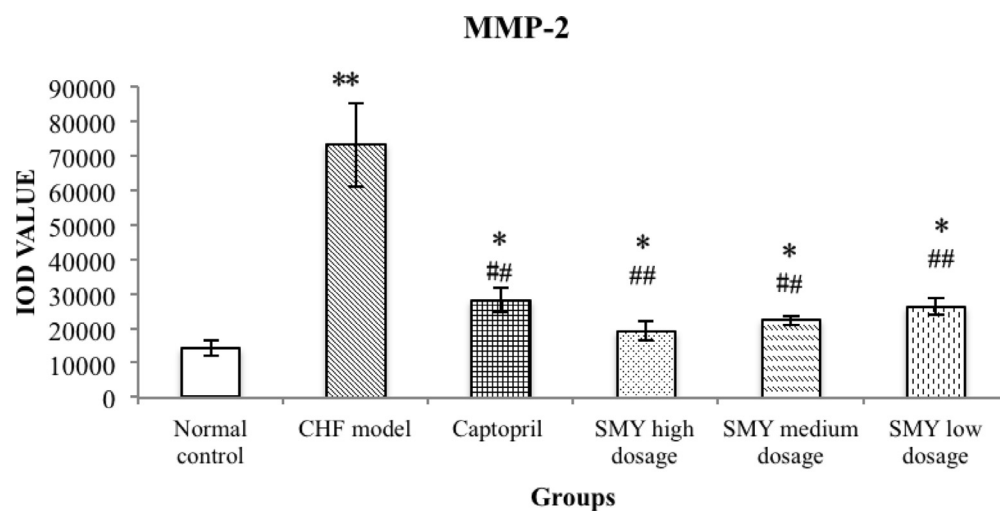


Fig. 7. Quantitative analysis of MMPs expression. IOD values represent the Mean $\pm$ SD. CON was the reference group in pairwise comparison. * indicated $p < .05$, ** indicated $p < .01$. CHF model group was the reference group in pairwise comparison. # indicated $p < .05$, ## indicated $p < .01$.

Fig. 8. Myocardial diameter (μm) among six groups (Mean $\pm$ SD). * indicates significant differences ($p < .05$) respect to CON, ** indicates extremely significant differences ($p < .01$) respect to CON, # indicates significant differences ($p < .05$) respect to CHF, ## indicates extremely significant differences ($p < .01$) respect to CHF.

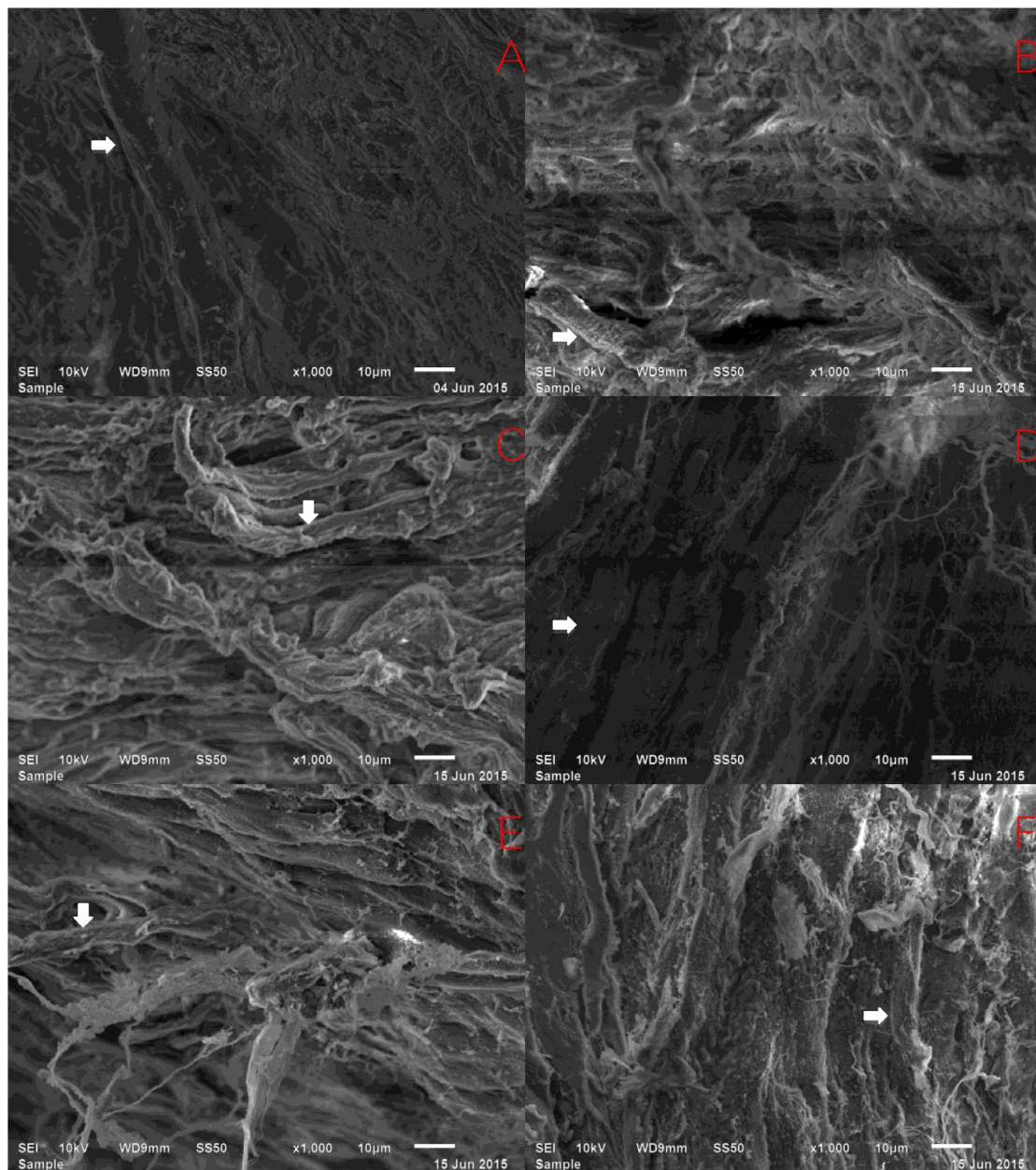


Fig. 9. SEM microphotographs ($\times 1000$) of myocardial fiber without any treatment (CON, A); B (CHFM): Adriamycin-induced was used to establish the rats model with CHF, and without any treatment. C (CA) was treated with Captopril, different dosage of SMY were administrated in rats model with CHF (HSMY, D; MSMY, E and LSMY, F).

research in effects of bioactive components on CHF progression.

In this study, The concentration of ginsenoside Rg1, ophiopogonin D, ophiopogonin D' and schisandrin in SMY was found to be 25.63 ± 3.42 mg, 11.00 ± 1.17 mg, 7.02 ± 0.51 mg, and 25.31 ± 4.28 mg per gram of SMY, respectively. TNF- α of four drug treated groups was significantly lower and IL-6 of CA, HSMY and MSMY dosage group was lower than CHFM. These observations suggest that CA and SMY have protective potential for the treatment of CHF. Thus SMY has similar treatment efficacy as captopril. Among the CA and the 3 SMY groups, no significant difference was observed ($p > .05$), however with decreasing dosage of SMY, both IL-6 and TNF- α showed and increasing trend ($p > .05$). This study suggests that owing to the lack of side effects, SMY can be used as a clinical treatment drug at high doses for the treatment of myocardial injury in patients with chronic heart failure.

Cardiac biomarkers reflecting different aspects of myocardial function, e.g., MR-pro ANP, NT-proBNP-atrial and ventricular wall stress

and troponin I (TNI)-myocyte injury, are increased upon injury (Gustafsson et al., 2009; Hillege et al., 2000), however it is not certain whether these biomarkers reflect decreased renal clearance or increased cardiac secretion (Bosselmann et al., 2013). Extracellular matrix (ECM) precipitation, with COL-IV as the major component, is the most characteristic pathological change of CHF (Zhao et al., 2014). Atrophic renal tubular epithelial cells are the main source of COL-IV expression in kidney (Ogata et al., 2002; Kimura et al., 2005). The results of present study proved that the protein expression and transcription level of COL-IV are consistent. Although, the increase in COL-IV expression could play a role in cardiac repair, the overexpression could also affect blood flow, and cause chronic ischemia and anoxic injury (Xie et al., 2009). On the other hand, under conditions of hypoxia and ischemia, epithelial-mesenchymal transdifferentiation would be favored, and lots of COL-IV would be synthesized by myofibroblasts. The myocardial injury modeling induced by adriamycin, led to muscle lesions, interstitial edema and interstitial fibrosis in the myocardium. IL-6 and TNF- α

are important assessment indexes for the myocardial injury, with excessive IL-6 and TNF- α promoting myocardial injury, and myocardial remodeling. Schisandrol A, an active component of SMY, has been reported to activate eNOS, lead to expression of Bcl-2, and reduce collagen deposition in myocardial tissues (Wang et al., 2010; Park et al., 2012). Our study indicated that SMY can reduce the levels of IL-6 and TNF- α ($p < 0.05$), and effectively attenuate the pathologic changes of myocardium, including myocardial fibrosis. COL-IV expression could be used as the main evaluation index for the assessment of degree of fibrosis of the cardiac tissues. COL-IV expression significantly decreased using high dose of SMY ($p < .05$). Moreover, with decreasing dosage of SMY, COL-IV expression showed an increasing trend ($p > .05$). These results illustrated a positive correlation between inflammatory cytokine (IL-6, TNF- α) levels and myocardial fibrosis index (COL-IV).

Myocardial fibrosis is a common pathological change in MI-induced heart failure as well as other end-stage cardiovascular diseases (Kong et al., 2014). It can induce detrimental left ventricular remodeling and eventually cardiovascular-related deaths (Eschaliere et al., 2013; Nguyen et al., 2014). Myocardial fibrosis is characterized by ECM overproduction of collagen in particular (Wang et al., 2005). MMPs are key enzymes for ECM degradation, which can degrade all the ECM components, except polysaccharides and play an important role in ECM degradation during myocardial interstitial remodeling. The expression of MMPs, ECM and collagen were significantly higher during heart failure, and the changes of MMPs regulate the degradation of collagen and ECM, which causes the cardiac/myocardial remodeling, and play a role in cardiac pathology process (Chen et al., 2004). Therefore, the overexpression of MMPs, ECM and collagen increase the risk of myocardial fibrosis (Huang, 2006). In the heart, MMP-1, MMP-2 and MMP-9 are the mainly expressed isoforms (Ulrich et al., 2004; Visse and Nagase, 2003). MMP-2 and MMP-9 are the limiting enzymes for collagen I (COL-I) and collagen III (COL-III) in the heart, that degrade the denatured collagen, gelatin, collagen-IV (COL-IV) and collagen V (COL-V), and regulate the speed of COL-I and COL-III degradation (Chen et al., 2004; López et al., 2004). Inhibition of MMPs could alleviate MI to some extent by improving the sensitivity of PPAR-[gamma] agonist (Chen et al., 2004). In this study, the results showed that expression of MMPs and collagen were significantly higher in SMY-treated groups as compared to the control group ($p < .01$), after the establishment of CHF model. This indicates that the changes in levels of MMPs, which are caused by CHF, can be reversed and thus treated using captopril and SMY, which effectively inhibit the expression of MMPs.

Conclusions

In conclusion, our data showed that Chinese herbal preparation SMY decreases the production of cytokine IL-6 and TNF- α to reduce the cardiac or myocardial injury. It also effectively attenuated or suppressed the pathological changes of myocardium on rats with CHF and inhibited the overproduction of MMP-2, MMP-9 and COL-IV, which significantly relieved cardiac pathology process, decreased the risk of myocardial fibrosis and cardiac remodeling. These findings suggest that SMY and captopril have the similar efficacy on adriamycin-induced myocardial injury, in addition, that Chinese herbal preparation SMY holds a vast potential in the treatment of cardiac diseases.

Authors' contributions

Z.K. and L.J. designed this study, including search strategy, eligibility criteria. Z.K., W.L. and P.M. selected studies and analyzed the data. P.A. and W.X. checked and repeated the data. Z.K. wrote the manuscript. All authors read and approved the final manuscript.

Conflict of interest

The authors do not have any conflict of interest.

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